

The background of the slide features a blurred image of two McLaren Formula 1 cars racing on a track. The cars are primarily blue and orange, with visible sponsor logos like 'Chromefibre' and 'McLaren'. The image is overlaid with a semi-transparent orange filter. On the left and right sides of the slide, there are large, solid orange rounded rectangular shapes.

INFS2608-GROUP3

ENHANCING MCLAREN F1 STRATEGY THROUGH DATA WAREHOUSING

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EXECUTIVE SUMMARY

Understanding the main objectives, a proof of concept illustrated through a specific problem statement, and the wider purpose of our database and data warehouse.

Objectives

ERD: Accurately reflect data that would be inputted in a real-life case study, and **maintain data integrity**

Data warehouse: Limit data redundancy to **optimise usability** for analysis through **Materialized Views & PowerBI**

Problem Statement

What are the steps to building a **proof of concept data warehouse** for the management for F1 drivers, which **captures all inflowing data accurately**, while eliminating **data redundancy** for **optimal usability** and analysis.

Proof of Concept

Scope focuses on the **direct data** from the management of F1 drivers; the **inputs** that go into their performance (wellbeing, training, race strategy, supporting staff etc), and their **output** (granular details about their race performance throughout the circuits).

Looking Forward



Update database to reflect real-world changes, **bolstering reliability and data integrity**



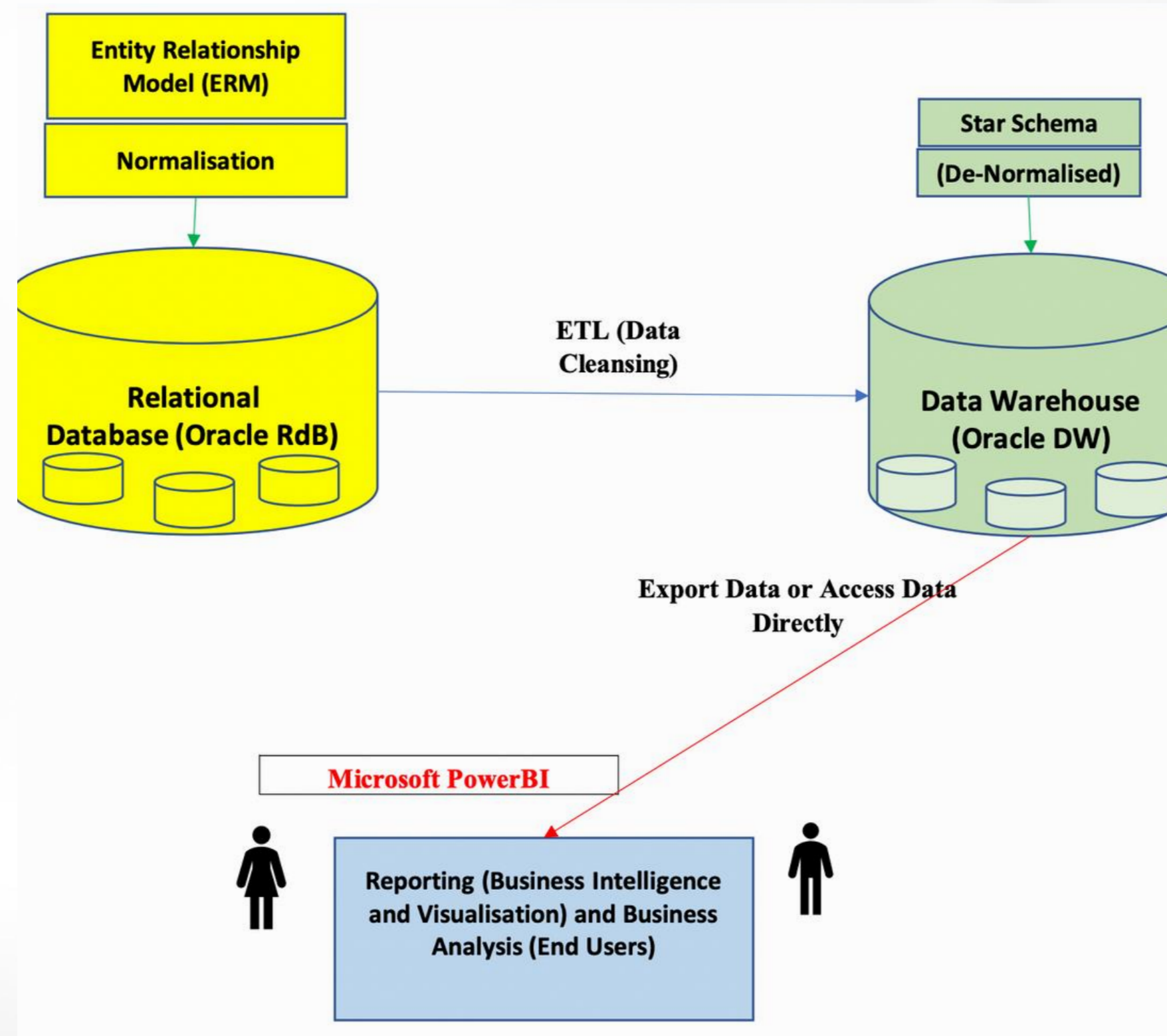
Expanding the scope of data warehouse while **maintaining intelligibility** for analysis

PROJECT FEASIBILITY

Our solution utilises **transaction data** that is captured directly from **F1 operations**, within the scope of our **database**.

The **operational data** goes through the crucial **ETL (extract, transform, load)** process, which **cleans the data, integrates it, and transforms it into a data warehouse**, ready to be utilised through **PowerBI visualisations**, for **end users and stakeholders**.

To understand if **PowerBI** will therefore be an **effective and efficient strategy**, we can look to a **case study (Walmart)** to understand the **implications of such a change to a much larger scenario**.



WALMART CASE STUDY

The goal of this project is to **democratize data**, to make **data for the management of F1 drivers** accessible and digestible to **not just those with technical backgrounds**.

This is exactly the same objective **Walmart's Finance Data & Analytics team** were tackling too. They used **PowerBI** as a **powerful, scalable and cost effective platform** that allowed analysts to **reduce the amount of time spent gathering data** and drill down to the granular level. Power BI **visualised insights into daily journal entries** from **Walmart transactions across the US**.

The data derived from the **management of F1 drivers** will be of **less volume and velocity** than of the **operations of Walmart**, a massive corporation.

Therefore, we believe through the **creation of a database**, which is then **transformed into a data warehouse**, and then **fed into PowerBI**, this project will be feasible for F1 teams to analyse the performance of their drivers.



Reusability
of data



Speed to
insight



No wasted
time



Centralization
of data

No duplicate
efforts



Easily
consumable



Automatic
refresh



RISKS



AND MITIGATIONS

The implementation of PowerBI as a new technology can lead to several **risks** that lead to **inefficiencies and poor data management**. PowerBI's **accessibility and integration** within the ubiquitous Microsoft ecosystem can present unique challenges.

- As PowerBI belongs integrates with the Microsoft ecosystem, it can be very appealing and intuitive to use. It can quickly become integrated into workflows **without proper oversight and governance, presenting a significant risk overnight**.
- **Poor data quality**: Ultimately, PowerBI uses data from the data warehouse, which covers **vast and various areas of F1 driver management**. Therefore, the quality of data stemming from these segments must be **verified** to ensure **visualisations are accurate and valid**.
- Members within the organisation **may lack the expertise** to utilise this new technology, or **feel unsupported or uncomfortable updating their current workflow**.

These risks can be **mitigated** to benefit from the advantages of PowerBI. While the ease in which PowerBI can be accessed and utilised is unique, many of these risks are present in the implementation of all new technologies at an organisation.

- **Ensuring that the right structures and oversights** are in place is vital in any form of risk management IT and analytics teams should work together to ensure that solutions are robust and scalable, and not unwieldy, unclear or unreliable.
- **Data quality profiling tools** can be use to assess the state of source data, and move towards improving its quality. Ultimately, working with the **most reliable data first** and **making it known** which data must be considered more warily limits mitigates this risk.
- For the database: **clear responsibilities** for data & database administrators. For end users, **consistent training & support** to inform them of its benefits will encourage adoption of PowerBI.

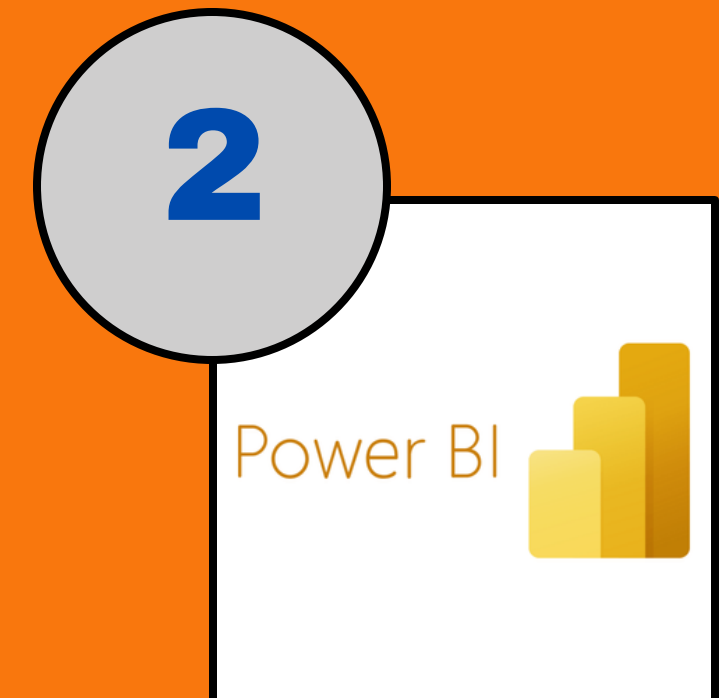
KEY FEATURES OF **OUR** **SOLUTION**

1. Scalable Big Data Handling

2. User Friendly Interface

3. Customizable Visuals

For example, during a live race weekend, the team can monitor **tire degradation and driver fatigue levels**, helping make **data-driven** pit stop decisions



PRACTICAL USE CASES



Driver Season Summary

→ Monitor wins, podiums, DNFs, overtakes, and average finishing positions to evaluate overall performance



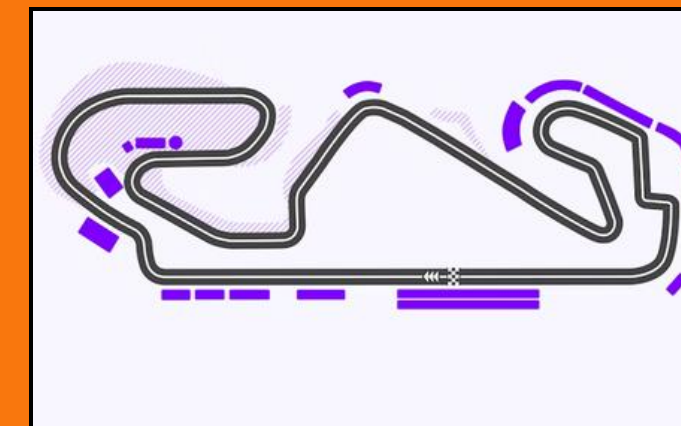
Driver Wellbeing vs Performance

→ Correlate stress levels and mental readiness with finishing positions to optimize driver training and support.



Race Strategy Impact

→ Analyze how tire selection, pit stop planning, and weather conditions affect race outcomes



Performance By Circuit

→ Identify which circuits suit each driver best, helping tailor car setup and race strategy

ANALYSING DATA FROM POWER BI

In F1, driver performance is critical, with each driver's **strengths varying by circuit**. PowerBI leverages **real-time adaptive data** to create **dashboards** that **evaluate drivers' skills and performance instantly**.

1

By understanding the **average position finishes** of drivers based on the **circuit**, the management team can **accurately delegate the correct driver for each race**.

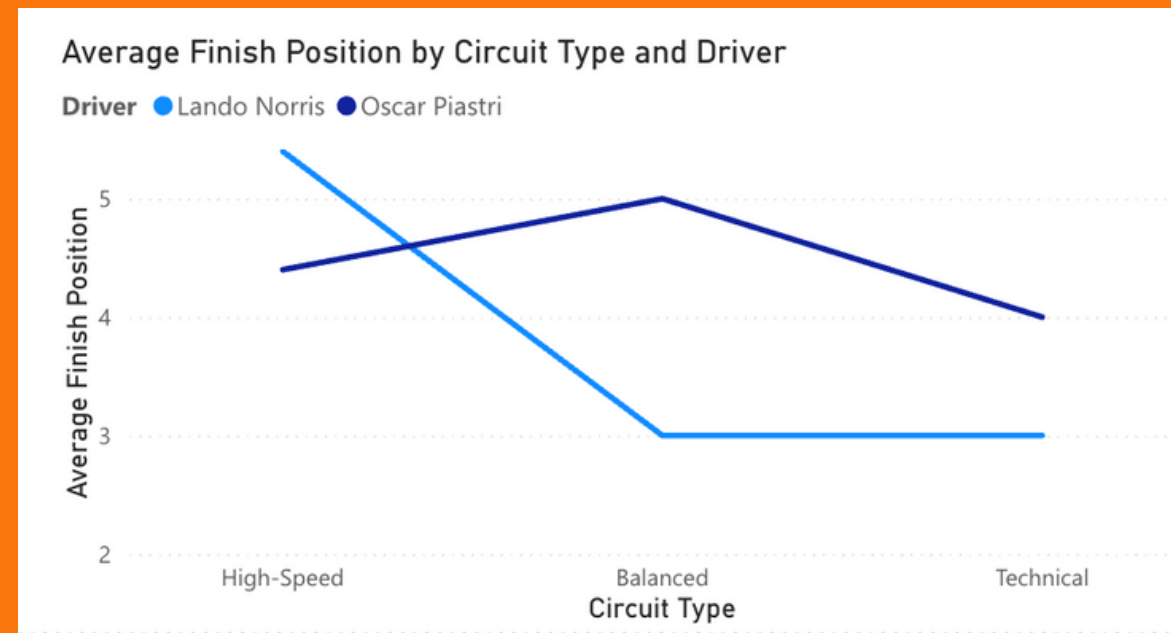
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Looking at **what circuit each driver performs the best at** provides teams the ability to **plan beforehand** and allow drivers to **play to their strengths**.

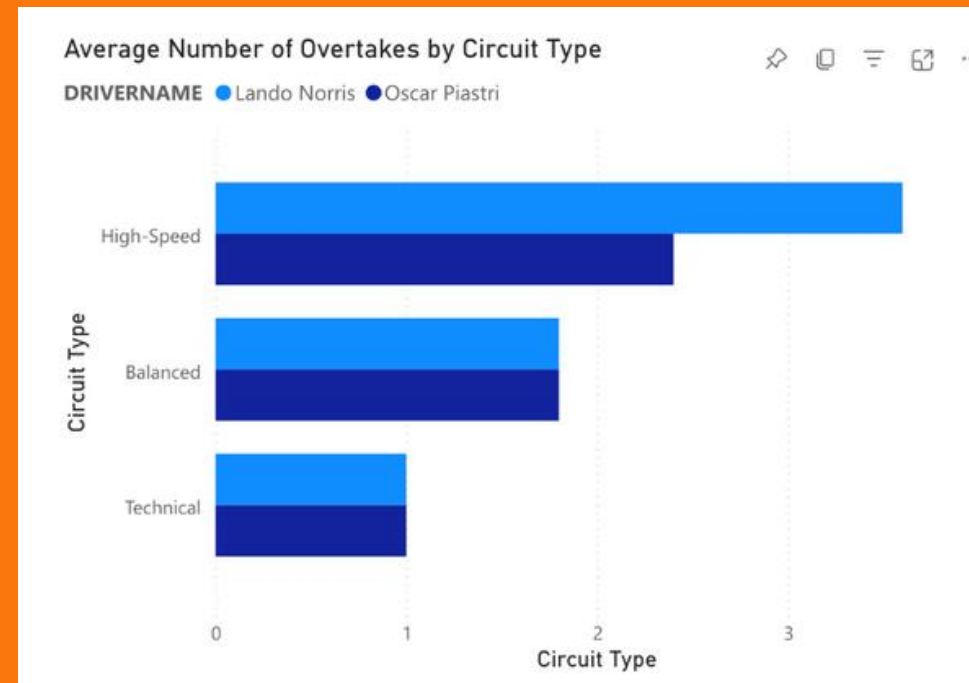
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Analysing factors for each drivers such as how the **circuit type impacts their tire degradation** allows an F1 managment team to plan accordingly.

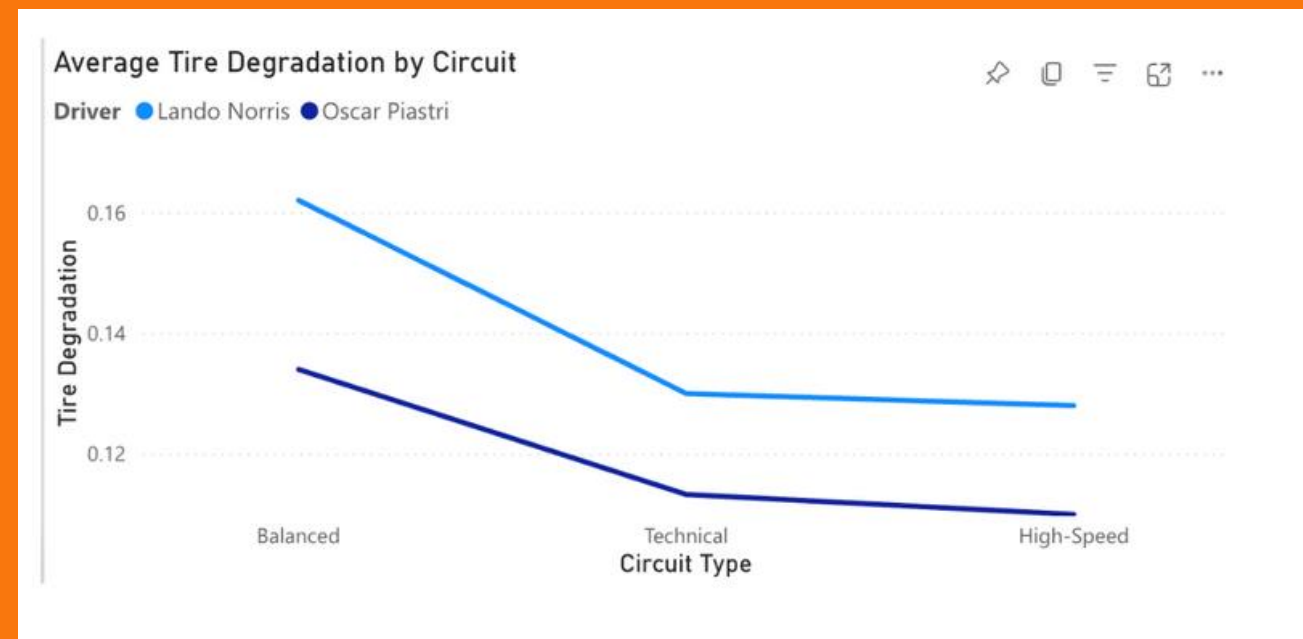
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PRESENTING **SOLUTION** **DEMONSTRATIONS**

Using both **PowerBI** and **SQL** provides an F1 management team with a **powerful, integrated approach** that allows them to **analyse driver performance** and **additionally wellbeing**.

Together, these tools **enhance decision-making, optimize driver development,** and support a **data-driven approach** to gaining **competitive advantage**.

USING **POWERBI**

Allows for

- **Dynamic visualisation of data**
- **Real-time analysis of data** which helps teams **identify patterns easily**
- Allows an F1 management team to **make informed decisions on strategy** and **training based on real-time data**

SQL BASED **DATABASE**

Allows for

- **Creation of a centralised data warehouse**
- Can easily **consolidate data** from **telemetry systems** and **training and performance logs**
- **Easy to manage and query** large volumes of **structured data efficiently**